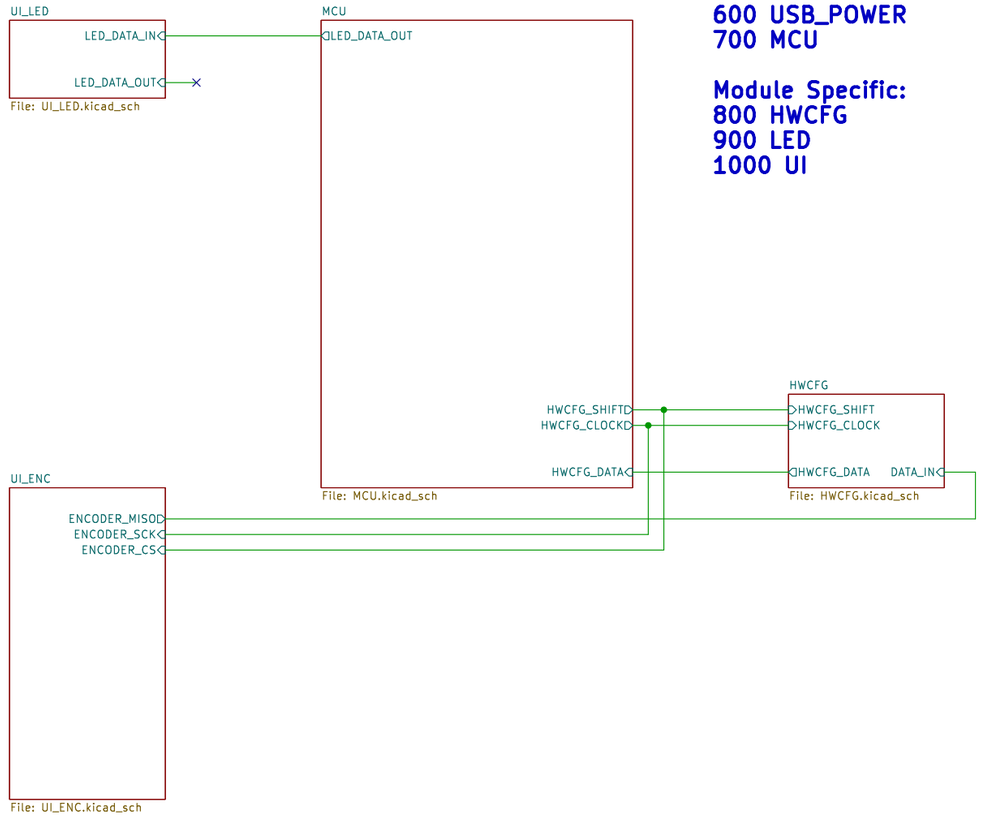


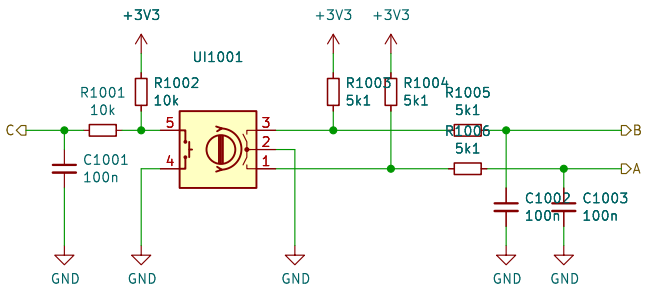
Common Sheets:
500 GRID
600 USB_POWER
700 MCU

Module Specific:
800 HWCFG
900 LED
1000 UI

Sheet: /
File: PCBA-EN16.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 8.0.8		Id: 1/23



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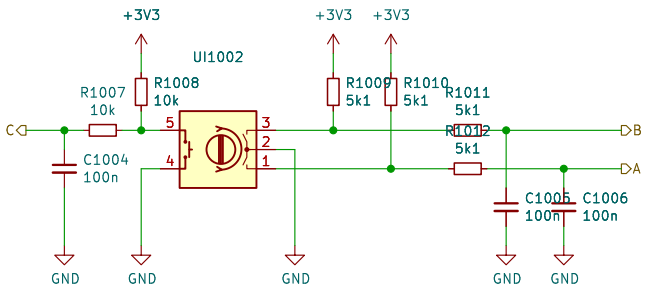
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 2/23



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File: UI_ENC_FILTER.kicad_sch

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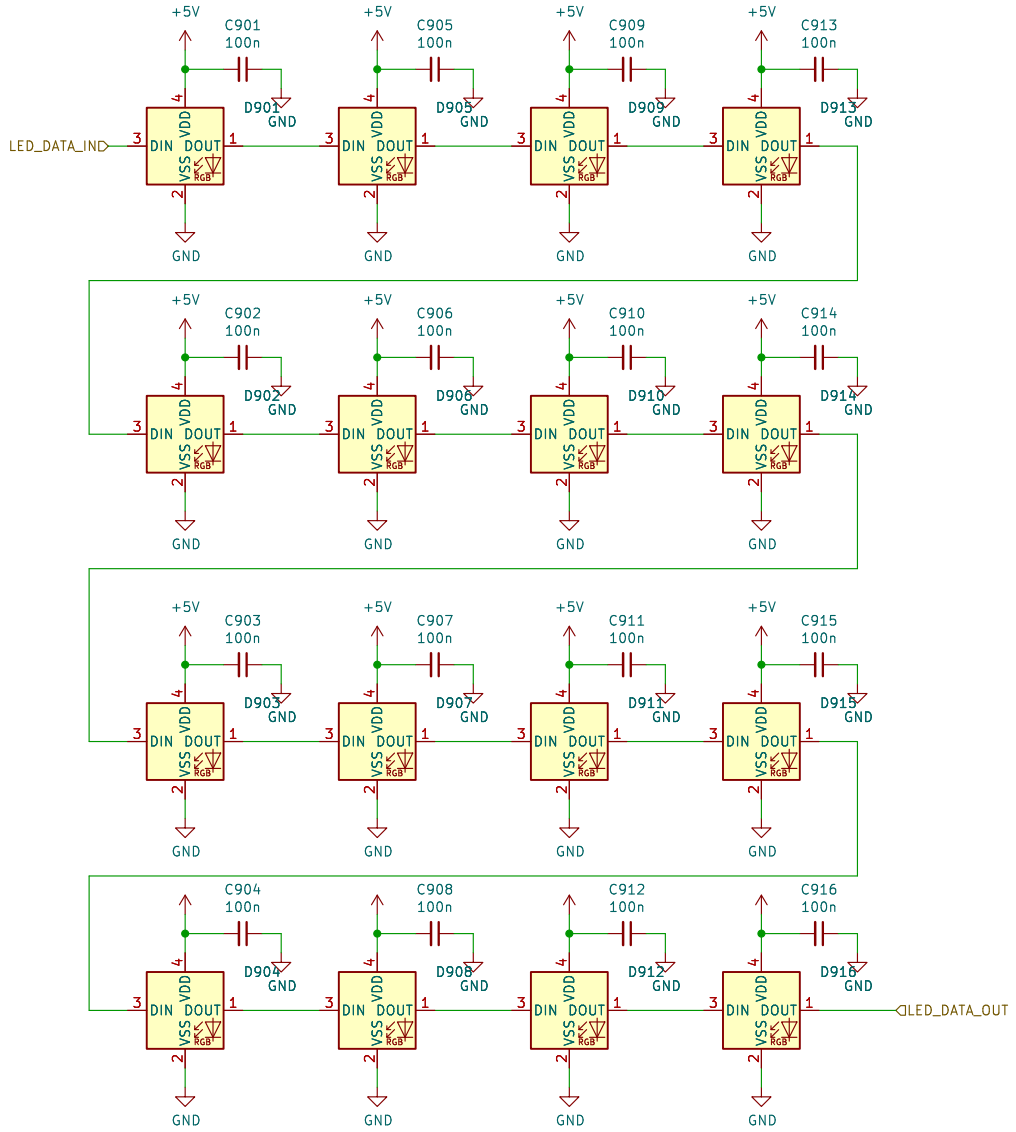
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 3/23

900



Sheet: /UI_LED/
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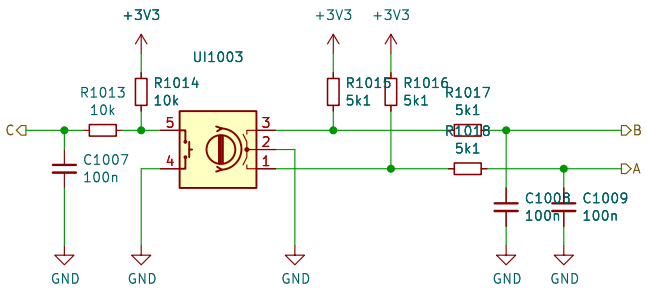
Size: A4	Date:	Rev:
KiCad E.D.A. 8.0.8		Id: 4/23



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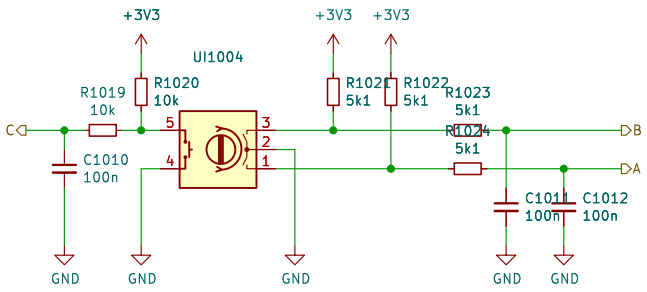
File: UI_ENC_FILTER.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 5/23



Sheet: /UI_ENC/ENCODER_12/
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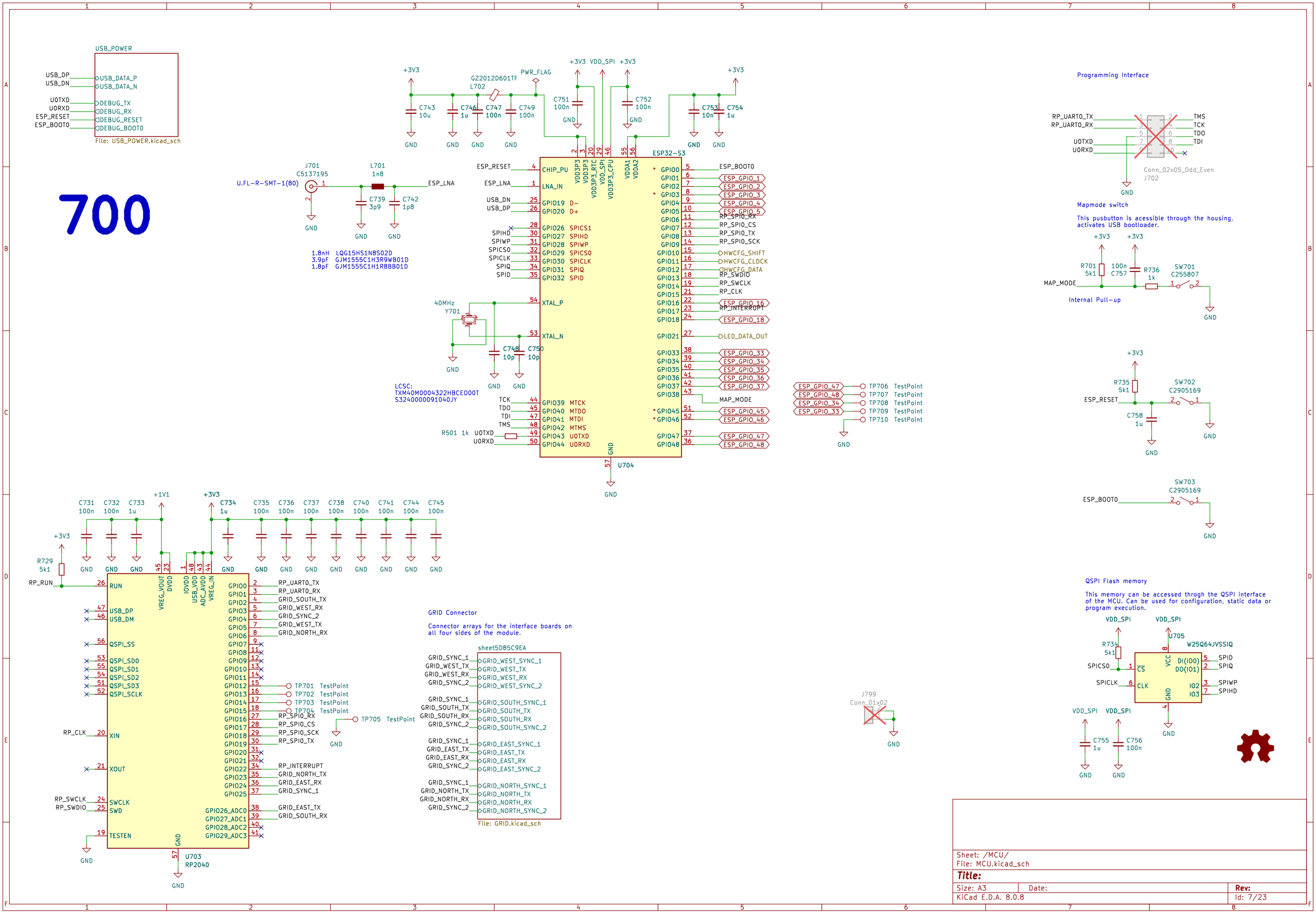
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 6/23

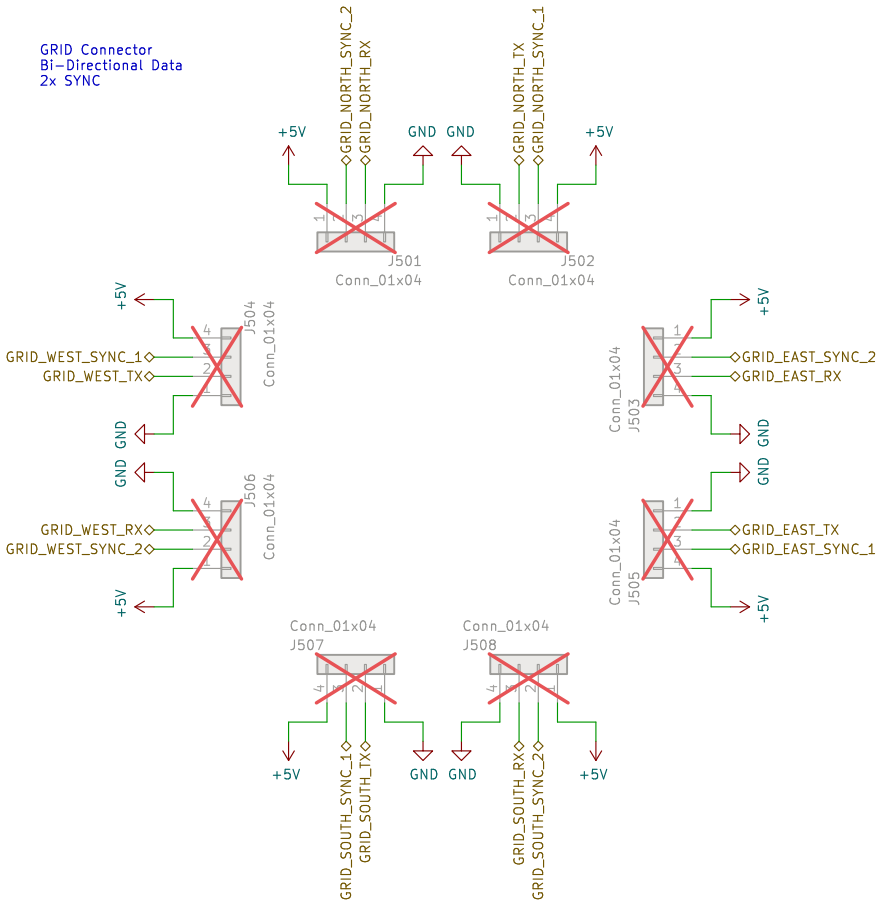


Sheet: /MCU/
File: MCU.kicad_sch

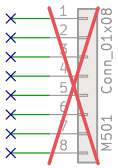
Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 8.0.8		Id: 7/23

GRID Connector
Bi-Directional Data
2x SYNC



Board Mounting Pattern



500

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File: GRID.kicad_sch

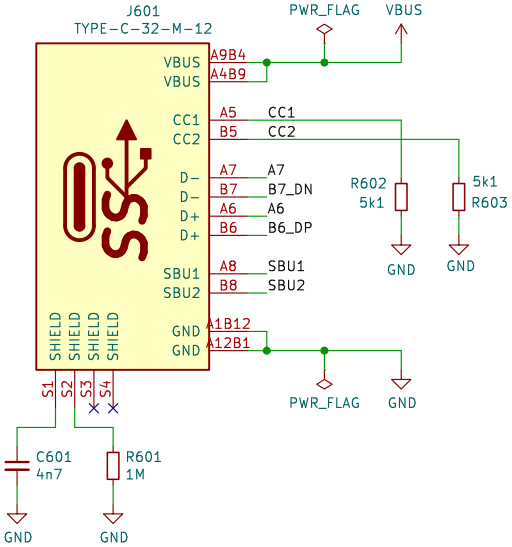
Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 8.0.8		Id: 8/23

600

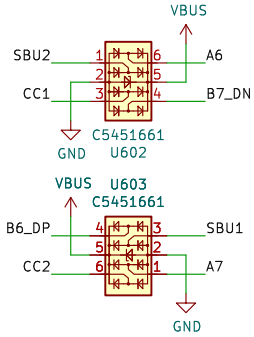
USB Port

USB C upstream facing port configured for 5V 3A power consumption.



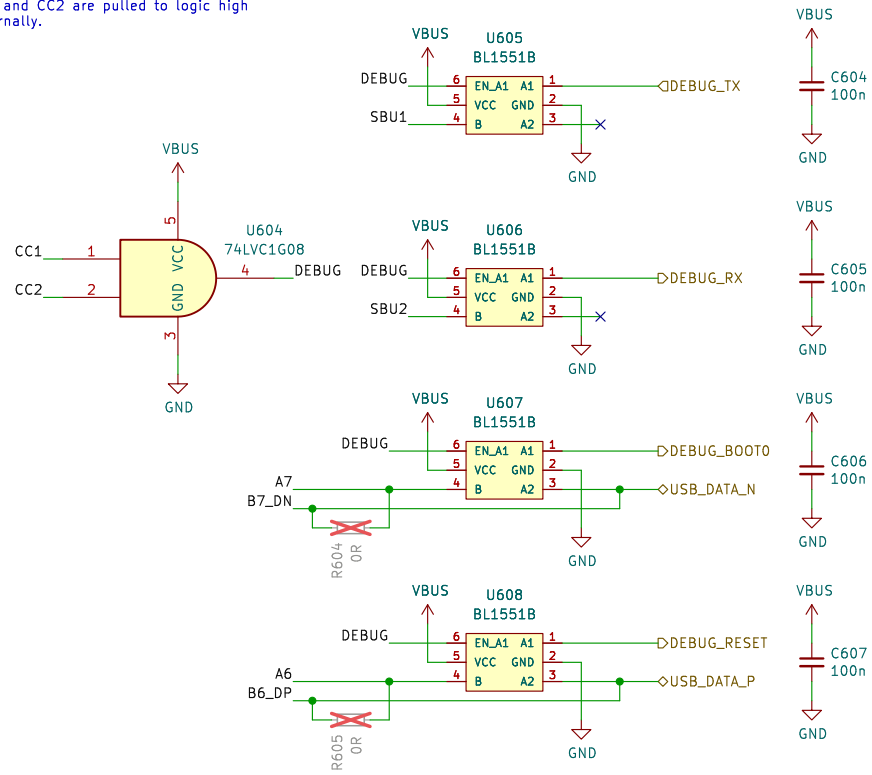
ESD Prot.

ESP protection for all 8 signals externally accessable via the USB C connector.



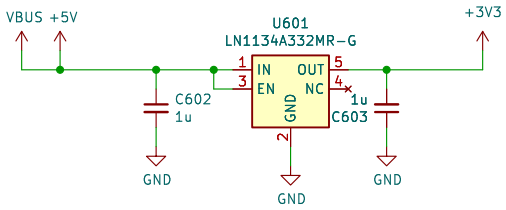
Debug-Mode Multiplexing

Debug.mode is activated when both CC1 and CC2 are pulled to logic high externally.

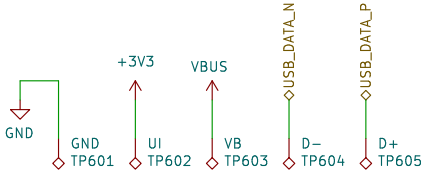


3V3 LDO

LDO regulator for generating the +3V3 power rail for the microcontroller and UI.



Testpoints

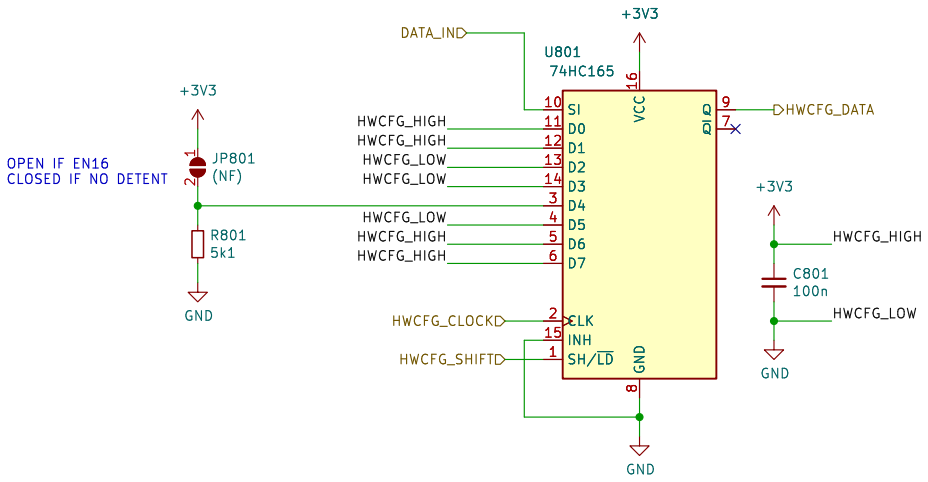


Sheet: /MCU/USB_POWER/
File: USB_POWER.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 8.0.8		Id: 9/23

800



Board Identification

Grid firmware can identify the hardware and the board revision thorugh a 3 wire serial interface using one or more shift register as read only memory. The content of the memory is defined by pulling the inputs high or low through pcb traces or solderable configuration jumpers.

4b'Model + 4b'Revision + nb'Reserved (Multiple shift registers)

D0: MODEL (LSB)
D1: MODEL
D2: MODEL
D3: MODEL (MSB)
D4: REVISION (LSB)
D5: REVISION
D6: REVISION
D7: REVISION (MSB)

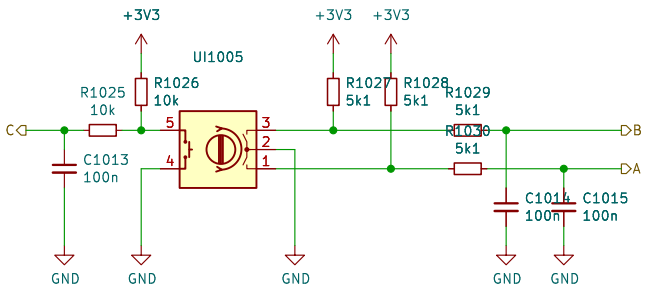
Model Codes (D3–D0):

Po16 0000
Bo16 0001
PBF4 0010
EN16 0011
...

Revision Codes (D7–D4):

RevA 0000
RevB 0001
RevC 0010
RevD 0011
...

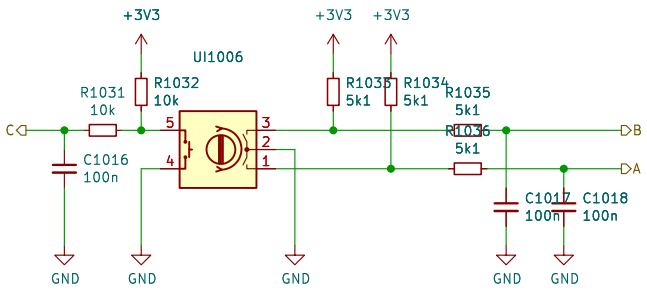
Sheet: /HWCFG/		
File: HWCFG.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 10/23



Sheet: /UI_ENC/ENCODER_0/
File: UI_ENC_FILTER.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 11/23



Sheet: /UI_ENC/ENCODER_6/
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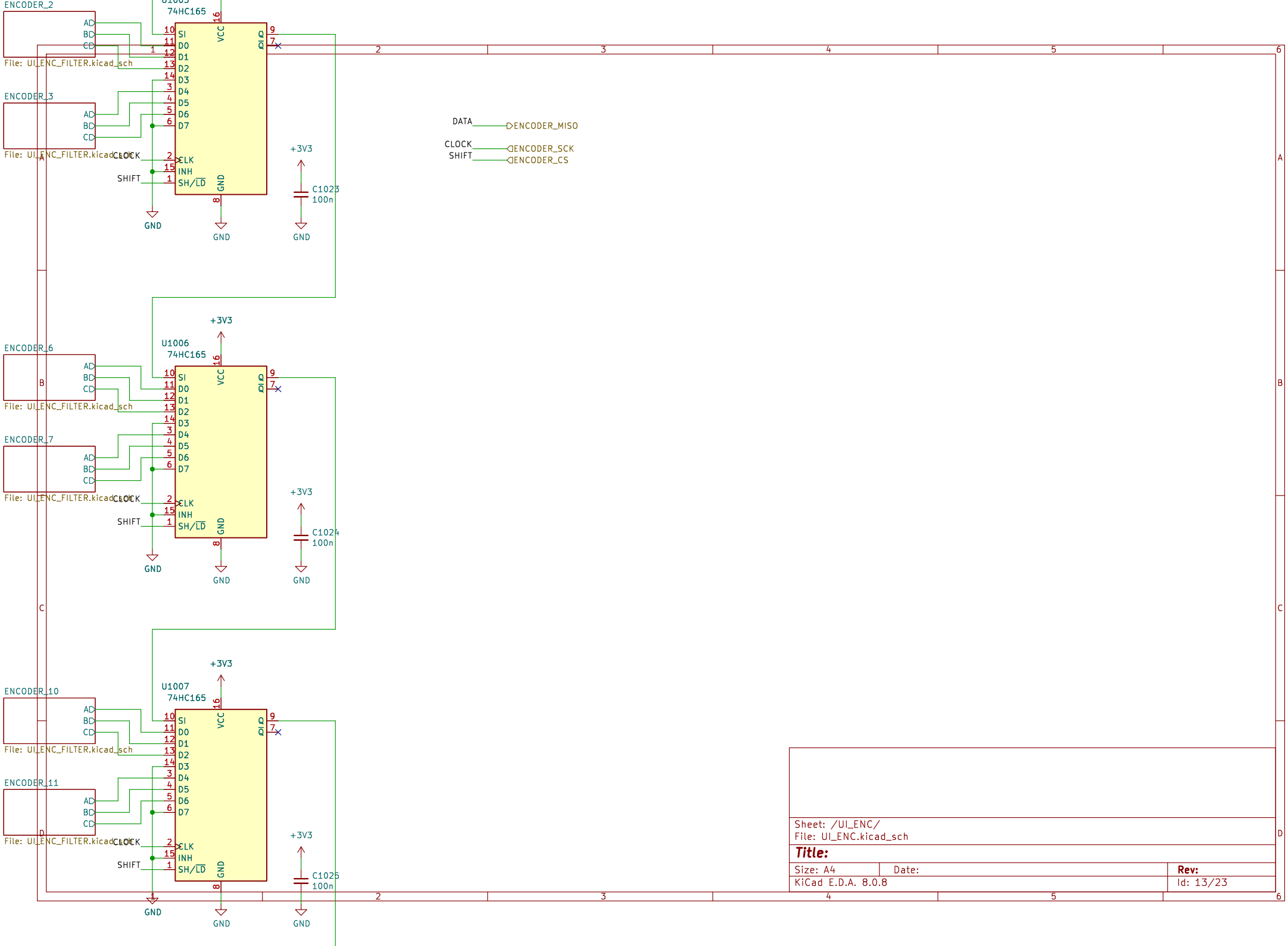
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 12/23


ENCODER_2
AD
BD
CD
File: UI_ENC_FILTER.kicad_sch
ENCODER_3
File: UI_ENC_FILTER.kicad_sch
U1005
74HC165
SI
D0
D1
D2
D3
D4
D5
D6
D7
CLK
INH
SH/LD
VCC
GND
Q7
CLOCK
SHIFT
+3V3
C1023
100n
DATA
ENCODER_MISO
CLOCK
ENCODER_SCK
SHIFT
ENCODER_CS
ENCODER_6
File: UI_ENC_FILTER.kicad_sch
ENCODER_7
File: UI_ENC_FILTER.kicad_sch
U1006
74HC165
C1024
100n
ENCODER_10
File: UI_ENC_FILTER.kicad_sch
ENCODER_11
File: UI_ENC_FILTER.kicad_sch
U1007
74HC165
C1025
100n
Sheet: /UI_ENC/
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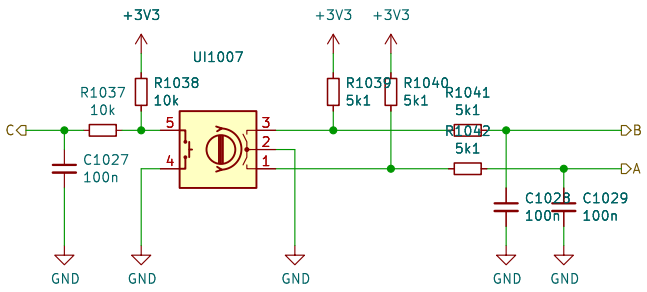
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.8
Id: 13/23



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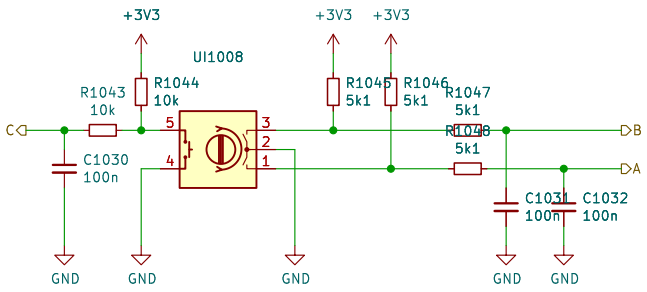
File: UI_ENC_FILTER.kicad_sch

Title:

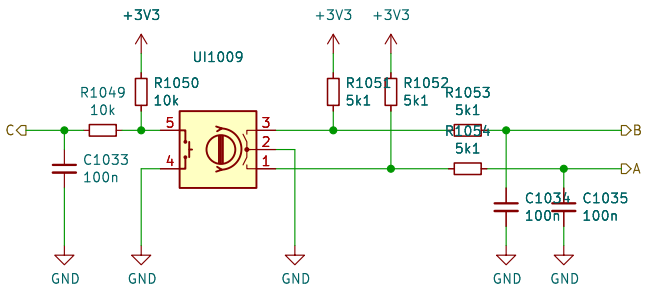
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 14/23



Sheet: /UI_ENC/ENCODER_1/
File: UI_ENC_FILTER.kicad_sch

Title:

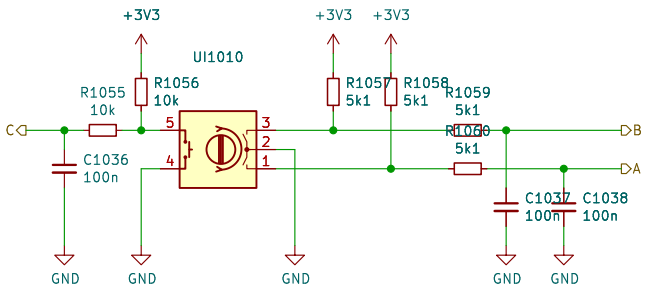
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 15/23



Sheet: /UI_ENC/ENCODER_7/
File: UI_ENC_FILTER.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 16/23



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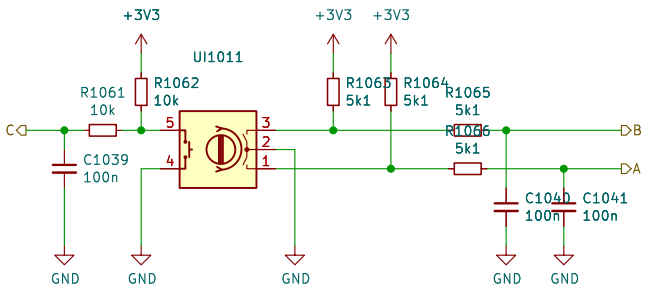
File: UI_ENC_FILTER.kicad_sch

Title:

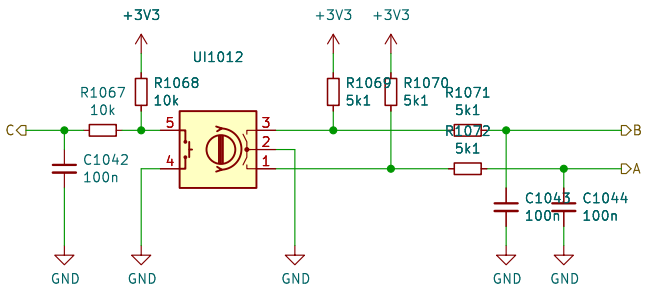
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 17/23



Sheet: /UI_ENC/ENCODER_13/
File: UI_ENC_FILTER.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 18/23



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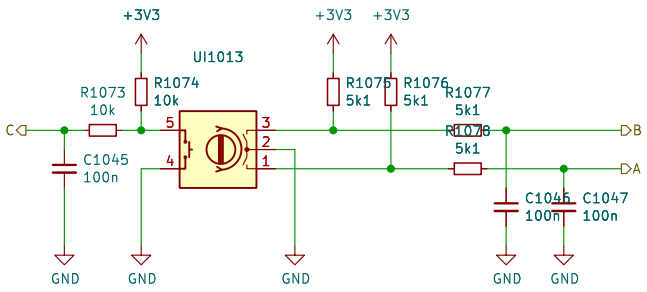
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 19/23



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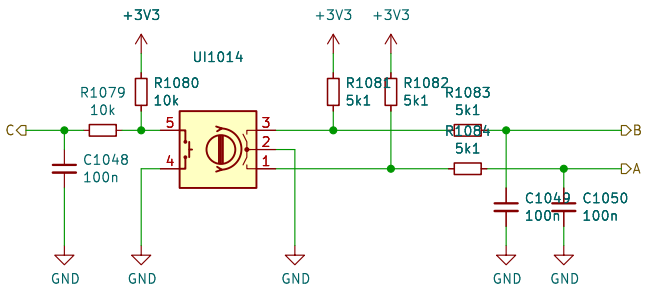
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 20/23



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File: UI_ENC_FILTER.kicad_sch

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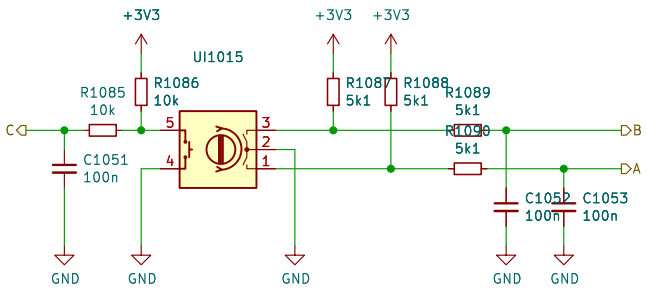
Size: A4	Date:	Rev:
KiCad E.D.A. 8.0.8		Id: 21/23



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File: UI_ENC_FILTER.kicad_sch

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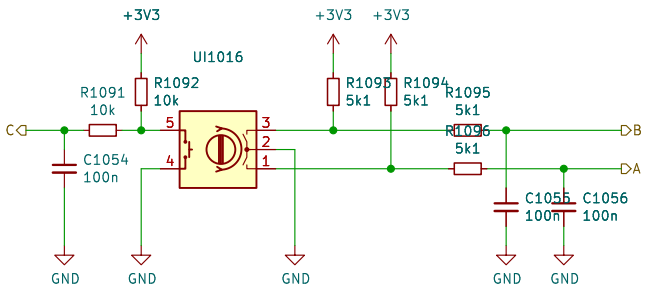
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 22/23



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File: UI_ENC_FILTER.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.8		Id: 23/23